

TABLE

DENOISING PERFORMANCE IN THE PRESENCE OF SIMULATED HEAD MOTION (TYPE 1). THE NOISE-FREE REFERENCE WITH THE MATCHED MOTION TRAJECTORY SERVED AS THE EVALUATION REFERENCE. RESULTS ARE REPORTED AS THE MEAN $\pm$ STANDARD DEVIATION ACROSS 30 INDEPENDENT REALIZATIONS. RED AND BLUE MEAN VALUES INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY, WITHIN EACH NOISE CONDITION. RANKINGS FOR $\Delta tSNR$ ARE BASED ON THE ABSOLUTE DISTANCE FROM 0.

(a) Gaussian

Noise	Method	MSE $\downarrow$	PSNR $\uparrow$	$\Delta tSNR \rightarrow 0$	FC_MAE $\downarrow$	FC_F $\downarrow$	r $\uparrow$
Gaussian=2	BM4D	1130.2 $\pm$ 0.2454	31.709 $\pm$ 0.0561	19.678 $\pm$ 0.4411	0.1365 $\pm$ 0.0035	17.673 $\pm$ 0.3759	0.8166 $\pm$ 0.0024
	MPPCA	0.2421 $\pm$ 0.0023	68.401 $\pm$ 0.0539	-0.2101 $\pm$ 0.0258	0.0028 $\pm$ 0.0001	0.4555 $\pm$ 0.0249	0.9948 $\pm$ 0.0002
	Patch2Self	0.7558 $\pm$ 0.0053	63.456 $\pm$ 0.0446	0.8315 $\pm$ 0.0831	0.0637 $\pm$ 0.0007	8.0844 $\pm$ 0.1020	0.9822 $\pm$ 0.0004
	VSTSVS	5.3197 $\pm$ 0.0893	54.982 $\pm$ 0.0549	0.0542 $\pm$ 0.0303	0.0053 $\pm$ 0.0003	0.7682 $\pm$ 0.0387	0.9954 $\pm$ 0.0003
	MD-S2S	60.982 $\pm$ 9.4651	44.389 $\pm$ 0.8493	19.225 $\pm$ 2.7758	0.3806 $\pm$ 0.0115	47.847 $\pm$ 1.3358	0.8327 $\pm$ 0.0076
	RMT-D	0.2324 $\pm$ 0.0017	68.579 $\pm$ 0.0481	0.1322 $\pm$ 0.0537	0.0036 $\pm$ 0.0002	0.7107 $\pm$ 0.0434	0.9953 $\pm$ 0.0001
Gaussian=5	BM4D	1139.5 $\pm$ 0.1940	31.673 $\pm$ 0.0562	14.512 $\pm$ 0.2801	0.1364 $\pm$ 0.0038	17.661 $\pm$ 0.4335	0.7879 $\pm$ 0.0030
	MPPCA	1.0548 $\pm$ 0.0049	62.009 $\pm$ 0.0490	-0.7023 $\pm$ 0.1058	0.0238 $\pm$ 0.0009	4.6352 $\pm$ 0.2077	0.9789 $\pm$ 0.0006
	Patch2Self	2.1215 $\pm$ 0.0203	58.974 $\pm$ 0.0478	1.4928 $\pm$ 0.1943	0.1300 $\pm$ 0.0016	15.849 $\pm$ 0.1802	0.9566 $\pm$ 0.0008
	VSTSVS	5.4227 $\pm$ 0.0495	54.899 $\pm$ 0.0453	-0.1133 $\pm$ 0.1131	0.0273 $\pm$ 0.0005	3.9239 $\pm$ 0.1108	0.9824 $\pm$ 0.0004
	MD-S2S	63.804 $\pm$ 9.0078	44.192 $\pm$ 0.6463	19.593 $\pm$ 1.3818	0.3867 $\pm$ 0.0109	47.445 $\pm$ 1.1003	0.8262 $\pm$ 0.0056
	RMT-D	1.0234 $\pm$ 0.0070	62.140 $\pm$ 0.0468	0.2181 $\pm$ 0.2072	0.0256 $\pm$ 0.0008	4.5821 $\pm$ 0.1210	0.9806 $\pm$ 0.0005
Gaussian=10	BM4D	1171.2 $\pm$ 0.2292	31.554 $\pm$ 0.0560	4.1778 $\pm$ 0.3739	0.1357 $\pm$ 0.0043	17.589 $\pm$ 0.5090	0.7258 $\pm$ 0.0038
	MPPCA	3.4144 $\pm$ 0.0282	56.908 $\pm$ 0.0617	-1.3374 $\pm$ 0.2324	0.1028 $\pm$ 0.0036	15.275 $\pm$ 0.6616	0.9430 $\pm$ 0.0012
	Patch2Self	4.2433 $\pm$ 0.0466	55.964 $\pm$ 0.0584	1.0265 $\pm$ 0.2998	0.2146 $\pm$ 0.0038	26.075 $\pm$ 0.4703	0.9244 $\pm$ 0.0014
	VSTSVS	6.9010 $\pm$ 0.0482	53.852 $\pm$ 0.0481	-0.6692 $\pm$ 0.2252	0.0842 $\pm$ 0.0014	11.451 $\pm$ 0.2367	0.9532 $\pm$ 0.0008
	MD-S2S	73.270 $\pm$ 5.5497	43.591 $\pm$ 0.8214	18.901 $\pm$ 1.4339	0.3740 $\pm$ 0.0201	46.220 $\pm$ 1.2372	0.8197 $\pm$ 0.0058
	RMT-D	3.0376 $\pm$ 0.0291	57.415 $\pm$ 0.0541	0.8919 $\pm$ 0.6449	0.0955 $\pm$ 0.0023	14.434 $\pm$ 0.3038	0.9481 $\pm$ 0.0010

(b) Rician

Noise	Method	MSE $\downarrow$	PSNR $\uparrow$	$\Delta tSNR \rightarrow 0$	FC_MAE $\downarrow$	FC_F $\downarrow$	r $\uparrow$
Rician=2	BM4D	1183.8 $\pm$ 0.4355	31.508 $\pm$ 0.0555	20.228 $\pm$ 0.4639	0.1445 $\pm$ 0.0032	18.687 $\pm$ 0.3463	0.8093 $\pm$ 0.0026
	MPPCA	0.3888 $\pm$ 0.0024	66.343 $\pm$ 0.0524	-0.2087 $\pm$ 0.0315	0.0027 $\pm$ 0.0001	0.4388 $\pm$ 0.0302	0.9946 $\pm$ 0.0002
	Patch2Self	0.9022 $\pm$ 0.0054	62.687 $\pm$ 0.0472	0.8263 $\pm$ 0.0790	0.0637 $\pm$ 0.0007	8.0683 $\pm$ 0.0914	0.9819 $\pm$ 0.0004
	VSTSVS	5.6667 $\pm$ 0.0608	54.707 $\pm$ 0.0543	0.0560 $\pm$ 0.0280	0.0052 $\pm$ 0.0004	0.7535 $\pm$ 0.0525	0.9952 $\pm$ 0.0003
	MD-S2S	56.580 $\pm$ 9.7013	44.755 $\pm$ 0.6687	17.637 $\pm$ 1.4605	0.3849 $\pm$ 0.0129	47.343 $\pm$ 1.3487	0.8256 $\pm$ 0.0053
	RMT-D	0.3787 $\pm$ 0.0018	66.458 $\pm$ 0.0523	0.1346 $\pm$ 0.0422	0.0036 $\pm$ 0.0003	0.6944 $\pm$ 0.0606	0.9951 $\pm$ 0.0001
Rician=5	BM4D	1193.3 $\pm$ 0.3564	31.473 $\pm$ 0.0555	15.494 $\pm$ 0.3101	0.1441 $\pm$ 0.0029	18.621 $\pm$ 0.2989	0.7826 $\pm$ 0.0031
	MPPCA	1.2094 $\pm$ 0.0060	61.415 $\pm$ 0.0548	-0.7048 $\pm$ 0.0999	0.0230 $\pm$ 0.0012	4.4269 $\pm$ 0.2745	0.9787 $\pm$ 0.0006
	Patch2Self	2.2758 $\pm$ 0.0209	58.669 $\pm$ 0.0492	1.4784 $\pm$ 0.1927	0.1298 $\pm$ 0.0016	15.812 $\pm$ 0.1887	0.9563 $\pm$ 0.0007
	VSTSVS	5.9558 $\pm$ 0.0224	54.491 $\pm$ 0.0554	-0.1112 $\pm$ 0.1045	0.0270 $\pm$ 0.0009	3.8775 $\pm$ 0.1927	0.9822 $\pm$ 0.0004
	MD-S2S	56.621 $\pm$ 7.0182	44.708 $\pm$ 0.6137	17.636 $\pm$ 2.2123	0.3891 $\pm$ 0.0151	47.669 $\pm$ 1.4190	0.8234 $\pm$ 0.0080
	RMT-D	1.1780 $\pm$ 0.0068	61.529 $\pm$ 0.0503	0.2207 $\pm$ 0.1647	0.0265 $\pm$ 0.0010	4.3973 $\pm$ 0.1461	0.9804 $\pm$ 0.0005
Rician=10	BM4D	1224.9 $\pm$ 0.3651	31.359 $\pm$ 0.0557	5.8514 $\pm$ 0.3606	0.1436 $\pm$ 0.0028	18.590 $\pm$ 0.2739	0.7241 $\pm$ 0.0038
	MPPCA	3.6178 $\pm$ 0.0296	56.656 $\pm$ 0.0606	-1.2629 $\pm$ 0.2408	0.1021 $\pm$ 0.0040	15.079 $\pm$ 0.6320	0.9427 $\pm$ 0.0010
	Patch2Self	4.4438 $\pm$ 0.0472	55.763 $\pm$ 0.0583	1.0038 $\pm$ 0.3135	0.2136 $\pm$ 0.0032	25.958 $\pm$ 0.4010	0.9241 $\pm$ 0.0012
	VSTSVS	8.1336 $\pm$ 0.0339	53.138 $\pm$ 0.0558	-0.6373 $\pm$ 0.2266	0.0834 $\pm$ 0.0017	11.332 $\pm$ 0.2395	0.9529 $\pm$ 0.0008
	MD-S2S	67.803 $\pm$ 9.5201	43.964 $\pm$ 0.5714	17.690 $\pm$ 2.5370	0.3886 $\pm$ 0.0117	47.654 $\pm$ 1.0436	0.8211 $\pm$ 0.0063
	RMT-D	3.2433 $\pm$ 0.0295	57.131 $\pm$ 0.0556	0.8994 $\pm$ 0.5020	0.0947 $\pm$ 0.0019	14.388 $\pm$ 0.2014	0.9479 $\pm$ 0.0009

TABLE

EFFECT OF PROCESSING ORDER UNDER SIMULATED HEAD MOTION. TYPE 2 DENOTES DENOISING FOLLOWED BY MOTION CORRECTION, AND TYPE 3 DENOTES MOTION CORRECTION FOLLOWED BY DENOISING. BOTH PIPELINES WERE EVALUATED AGAINST THE NOISE-FREE, MOTION-FREE GT. RESULTS ARE REPORTED AS THE MEAN $\pm$ STANDARD DEVIATION ACROSS 30 INDEPENDENT REALIZATIONS. RED AND BLUE MEAN VALUES INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY, WITHIN EACH NOISE CONDITION. RANKINGS FOR Δ tSNR ARE BASED ON THE ABSOLUTE DISTANCE FROM 0.

(a) Gaussian

Noise	Method	Type	MSE↓	PSNR↑	Δ tSNR→0	FC_MAE↓	FC_F↓	r↑
Gaussian=2	BM4D	2	1170.6 $\pm$ 0.2391	31.185 $\pm$ 0.0297	-23.339 $\pm$ 1.7514	0.1824 $\pm$ 0.0054	21.547 $\pm$ 0.5976	0.7764 $\pm$ 0.0041
		3	1154.8 $\pm$ 0.3270	31.244 $\pm$ 0.0293	-21.834 $\pm$ 1.7391	0.1824 $\pm$ 0.0058	21.539 $\pm$ 0.6415	0.7868 $\pm$ 0.0040
	MPPCA	2	20.505 $\pm$ 0.1049	48.751 $\pm$ 0.0462	-41.464 $\pm$ 2.1435	0.0143 $\pm$ 0.0009	2.2813 $\pm$ 0.1904	0.7313 $\pm$ 0.0055
		3	20.393 $\pm$ 0.1043	48.774 $\pm$ 0.0461	-40.855 $\pm$ 2.0973	0.0133 $\pm$ 0.0006	2.2135 $\pm$ 0.1883	0.7353 $\pm$ 0.0054
	Patch2Self	2	20.988 $\pm$ 0.1053	48.650 $\pm$ 0.0456	-38.868 $\pm$ 2.1740	0.0842 $\pm$ 0.0016	10.153 $\pm$ 0.1741	0.7032 $\pm$ 0.0059
		3	20.970 $\pm$ 0.1041	48.653 $\pm$ 0.0452	-38.857 $\pm$ 2.1753	0.0829 $\pm$ 0.0015	10.026 $\pm$ 0.1680	0.7039 $\pm$ 0.0059
	VSTSVS	2	26.661 $\pm$ 0.0962	47.610 $\pm$ 0.0377	-41.592 $\pm$ 2.1417	0.0190 $\pm$ 0.0010	3.0961 $\pm$ 0.1620	0.7288 $\pm$ 0.0054
		3	25.243 $\pm$ 0.0919	47.848 $\pm$ 0.0377	-40.351 $\pm$ 2.0887	0.0150 $\pm$ 0.0009	2.3043 $\pm$ 0.1766	0.7359 $\pm$ 0.0053
	MD-S2S	2	83.173 $\pm$ 9.0969	42.671 $\pm$ 0.8425	-33.982 $\pm$ 2.7399	0.6591 $\pm$ 0.0155	69.429 $\pm$ 1.1752	0.5027 $\pm$ 0.0052
		3	50.249 $\pm$ 9.5680	44.937 $\pm$ 0.8463	-19.335 $\pm$ 2.6928	0.6114 $\pm$ 0.0127	68.903 $\pm$ 1.2852	0.5945 $\pm$ 0.0096
	RMT-D	2	20.506 $\pm$ 0.1050	48.750 $\pm$ 0.0462	-39.086 $\pm$ 2.0318	0.0182 $\pm$ 0.0009	3.0280 $\pm$ 0.1298	0.7270 $\pm$ 0.0056
		3	20.306 $\pm$ 0.1044	48.793 $\pm$ 0.0461	-38.330 $\pm$ 1.9823	0.0142 $\pm$ 0.0008	2.2714 $\pm$ 0.1449	0.7318 $\pm$ 0.0055
Gaussian=5	BM4D	2	1180.1 $\pm$ 0.2018	31.150 $\pm$ 0.0299	-32.731 $\pm$ 2.2468	0.1814 $\pm$ 0.0055	21.415 $\pm$ 0.6102	0.7301 $\pm$ 0.0051
		3	1163.8 $\pm$ 0.2367	31.210 $\pm$ 0.0298	-31.548 $\pm$ 2.2302	0.1813 $\pm$ 0.0058	21.424 $\pm$ 0.6500	0.7388 $\pm$ 0.0050
	MPPCA	2	21.305 $\pm$ 0.1037	48.584 $\pm$ 0.0453	-43.885 $\pm$ 2.3142	0.0395 $\pm$ 0.0016	6.5538 $\pm$ 0.2808	0.7050 $\pm$ 0.0061
		3	20.894 $\pm$ 0.1010	48.669 $\pm$ 0.0450	-42.024 $\pm$ 2.2153	0.0355 $\pm$ 0.0012	6.0955 $\pm$ 0.2700	0.7125 $\pm$ 0.0059
	Patch2Self	2	22.306 $\pm$ 0.1059	48.385 $\pm$ 0.0439	-37.424 $\pm$ 2.3274	0.1844 $\pm$ 0.0032	21.017 $\pm$ 0.3331	0.6544 $\pm$ 0.0066
		3	22.116 $\pm$ 0.1021	48.422 $\pm$ 0.0431	-37.159 $\pm$ 2.3317	0.1814 $\pm$ 0.0028	20.696 $\pm$ 0.2959	0.6568 $\pm$ 0.0064
	VSTSVS	2	26.791 $\pm$ 0.0963	47.589 $\pm$ 0.0416	-43.342 $\pm$ 2.3148	0.0442 $\pm$ 0.0020	5.8752 $\pm$ 0.2432	0.7066 $\pm$ 0.0059
		3	24.960 $\pm$ 0.0912	47.897 $\pm$ 0.0420	-40.356 $\pm$ 2.1947	0.0390 $\pm$ 0.0014	5.2111 $\pm$ 0.1821	0.7228 $\pm$ 0.0057
	MD-S2S	2	83.697 $\pm$ 5.0950	42.643 $\pm$ 0.4715	-36.321 $\pm$ 3.1296	0.6153 $\pm$ 0.0079	69.100 $\pm$ 0.9065	0.5074 $\pm$ 0.0061
		3	51.938 $\pm$ 4.8237	44.731 $\pm$ 0.3788	-17.082 $\pm$ 3.0995	0.6050 $\pm$ 0.0083	68.254 $\pm$ 0.9171	0.6004 $\pm$ 0.0064
	RMT-D	2	21.297 $\pm$ 0.1028	48.586 $\pm$ 0.0451	-33.666 $\pm$ 1.8749	0.0426 $\pm$ 0.0022	5.8517 $\pm$ 0.2536	0.6883 $\pm$ 0.0066
		3	20.560 $\pm$ 0.1002	48.739 $\pm$ 0.0449	-30.658 $\pm$ 1.7153	0.0321 $\pm$ 0.0062	5.8312 $\pm$ 0.8151	0.7166 $\pm$ 0.0064
Gaussian=10	BM4D	2	1222.5 $\pm$ 0.2008	30.997 $\pm$ 0.0295	-49.749 $\pm$ 2.9281	0.1762 $\pm$ 0.0059	20.969 $\pm$ 0.6834	0.6424 $\pm$ 0.0065
		3	1205.0 $\pm$ 0.2453	31.059 $\pm$ 0.0292	-49.044 $\pm$ 2.9207	0.1760 $\pm$ 0.0064	20.942 $\pm$ 0.7340	0.6482 $\pm$ 0.0065
	MPPCA	2	23.826 $\pm$ 0.1112	48.099 $\pm$ 0.0427	-47.190 $\pm$ 2.5149	0.1563 $\pm$ 0.0067	21.957 $\pm$ 1.1529	0.6463 $\pm$ 0.0069
		3	22.776 $\pm$ 0.1021	48.295 $\pm$ 0.0426	-41.399 $\pm$ 2.2565	0.1512 $\pm$ 0.0060	21.417 $\pm$ 0.9812	0.6744 $\pm$ 0.0065
	Patch2Self	2	24.533 $\pm$ 0.1159	47.972 $\pm$ 0.0430	-37.995 $\pm$ 2.4847	0.3335 $\pm$ 0.0049	37.050 $\pm$ 0.5716	0.6001 $\pm$ 0.0068
		3	23.953 $\pm$ 0.1087	48.076 $\pm$ 0.0418	-37.094 $\pm$ 2.4903	0.3289 $\pm$ 0.0048	36.552 $\pm$ 0.5388	0.6048 $\pm$ 0.0065
	VSTSVS	2	28.473 $\pm$ 0.1043	47.325 $\pm$ 0.0412	-47.008 $\pm$ 2.5337	0.1284 $\pm$ 0.0035	16.413 $\pm$ 0.4973	0.6606 $\pm$ 0.0066
		3	25.536 $\pm$ 0.0961	47.798 $\pm$ 0.0421	-40.608 $\pm$ 2.3128	0.1250 $\pm$ 0.0027	16.122 $\pm$ 0.4900	0.6928 $\pm$ 0.0062
	MD-S2S	2	88.058 $\pm$ 5.0947	42.424 $\pm$ 0.3996	-33.980 $\pm$ 3.2549	0.6015 $\pm$ 0.0082	68.670 $\pm$ 0.8287	0.5025 $\pm$ 0.0063
		3	67.891 $\pm$ 5.7737	43.565 $\pm$ 0.3694	-15.280 $\pm$ 3.9963	0.5970 $\pm$ 0.0079	67.538 $\pm$ 0.8395	0.5970 $\pm$ 0.0069
	RMT-D	2	23.488 $\pm$ 0.1039	48.161 $\pm$ 0.0430	-21.785 $\pm$ 1.5609	0.1596 $\pm$ 0.0049	21.453 $\pm$ 0.6533	0.6094 $\pm$ 0.0082
		3	21.364 $\pm$ 0.0958	48.572 $\pm$ 0.0429	-19.900 $\pm$ 1.0474	0.1499 $\pm$ 0.0046	18.812 $\pm$ 0.7335	0.6420 $\pm$ 0.0079

(b) Rician

Noise	Method	Type	MSE↓	PSNR↑	AtSNR→0	FC_MAE↓	FC_F↓	r↑
Rician=2	BM4D	2	1224.8±0.3659	30.988±0.0297	-24.265±1.8379	0.1905±0.0061	22.532±0.6141	0.7678±0.0045
		3	1211.2±0.3558	31.037±0.0299	-23.616±1.7533	0.1899±0.0066	22.583±0.6868	0.7730±0.0056
	MPPCA	2	20.638±0.1063	48.722±0.0463	-41.461±2.1520	0.0143±0.0009	2.2838±0.1989	0.7311±0.0054
		3	20.527±0.1063	48.746±0.0462	-40.855±2.1026	0.0134±0.0009	2.2174±0.1973	0.7351±0.0053
	Patch2Self	2	21.122±0.1064	48.622±0.0456	-38.867±2.1762	0.0842±0.0016	10.153±0.1794	0.7030±0.0059
		3	21.102±0.1064	48.626±0.0454	-38.857±2.1779	0.0829±0.0015	10.025±0.1735	0.7038±0.0059
	VSTSVS	2	26.937±0.0947	47.566±0.0365	-41.567±2.1447	0.0189±0.0010	3.0854±0.1718	0.7288±0.0054
		3	25.542±0.0903	47.797±0.0374	-40.355±2.0928	0.0150±0.0010	2.3056±0.1780	0.7357±0.0053
	MD-S2S	2	80.477±5.0983	42.810±0.4026	-38.805±3.6873	0.6041±0.0065	67.742±0.8130	0.4975±0.0092
		3	49.899±5.0934	44.908±0.4174	-18.113±3.9234	0.5951±0.0088	67.285±0.9119	0.5980±0.0097
	RMT-D	2	20.640±0.1069	48.722±0.0464	-39.080±2.0322	0.0182±0.0008	3.0216±0.1245	0.7268±0.0056
		3	20.440±0.1066	48.764±0.0463	-38.332±1.9817	0.0147±0.0017	2.2401±0.2446	0.7316±0.0055
Rician=5	BM4D	2	1234.4±0.2510	30.955±0.0294	-32.780±2.2662	0.1889±0.0063	22.342±0.6354	0.7256±0.0053
		3	1220.3±0.2994	31.005±0.0300	-32.156±2.2011	0.1885±0.0067	22.411±0.6982	0.7310±0.0064
	MPPCA	2	21.441±0.1081	48.557±0.0457	-43.885±2.3312	0.0382±0.0020	6.2744±0.3909	0.7048±0.0060
		3	21.034±0.1049	48.640±0.0452	-42.043±2.2307	0.0339±0.0019	5.7265±0.4034	0.7163±0.0058
	Patch2Self	2	22.444±0.1096	48.358±0.0444	-37.430±2.3315	0.1846±0.0033	21.021±0.3844	0.6542±0.0065
		3	22.252±0.1070	48.396±0.0437	-37.172±2.3376	0.1515±0.0028	20.693±0.3269	0.6565±0.0064
	VSTSVS	2	27.254±0.1093	47.515±0.0420	-43.311±2.3190	0.0440±0.0018	5.8494±0.2467	0.7065±0.0059
		3	25.459±0.1051	47.811±0.0428	-40.368±2.2030	0.0387±0.0013	5.1601±0.2060	0.7224±0.0056
	MD-S2S	2	90.083±8.0931	42.321±0.0420	-34.748±2.1447	0.6127±0.0070	68.700±0.7596	0.5212±0.0080
		3	54.649±8.3226	44.539±0.6644	-17.444±2.8490	0.6080±0.0079	68.540±0.7898	0.5996±0.0082
	RMT-D	2	21.431±0.1089	48.559±0.0460	-33.654±1.8778	0.0523±0.0020	6.7340±0.2275	0.6882±0.0065
		3	20.700±0.1048	48.710±0.0455	-30.663±1.6997	0.0299±0.0014	5.0831±0.2257	0.7022±0.0063
Rician=10	BM4D	2	1277.5±0.4465	30.806±0.0300	-48.586±2.8930	0.1833±0.0061	21.847±0.6374	0.6439±0.0064
		3	1262.2±0.4100	30.858±0.0302	-48.154±2.8616	0.1836±0.0065	21.991±0.6963	0.6485±0.0076
	MPPCA	2	24.016±0.1177	48.064±0.0437	-47.086±2.5483	0.1537±0.0078	21.516±1.1473	0.6459±0.0069
		3	22.973±0.1131	48.257±0.0441	-41.314±2.3455	0.1488±0.0066	21.024±1.0192	0.6740±0.0066
	Patch2Self	2	24.726±0.1224	47.938±0.0435	-38.009±2.5080	0.3316±0.0052	36.861±0.6172	0.5997±0.0069
		3	24.146±0.1156	48.041±0.0426	-37.131±2.5169	0.3270±0.0047	36.354±0.5372	0.6043±0.0065
	VSTSVS	2	29.649±0.1226	47.149±0.0417	-46.951±2.5569	0.1261±0.0037	16.139±0.4619	0.6605±0.0066
		3	26.741±0.1097	47.597±0.0424	-40.587±2.3582	0.1226±0.0035	15.807±0.4846	0.6924±0.0062
	MD-S2S	2	89.763±9.1128	42.341±0.0402	-33.929±3.2077	0.6182±0.0076	68.165±0.7864	0.5156±0.0083
		3	67.132±9.2340	43.651±0.7347	-20.632±3.4119	0.6036±0.0085	68.072±0.9717	0.6002±0.0093
	RMT-D	2	23.678±0.1149	48.126±0.0447	-21.737±1.6070	0.1515±0.0044	21.148±0.4537	0.6093±0.0081
		3	21.571±0.1050	48.531±0.0444	-19.996±1.0381	0.1469±0.0034	18.344±0.3635	0.6416±0.0080